

Information Retrieval

12. Wrap-up

Ghislain Fourny
Spring 2022

12

Lecture Overview

Introduction

Boolean queries

Term vocabulary and posting lists

Tolerant retrieval

Basics of Information Retrieval

Evaluation

Scale up

Index compression

Advanced topics

Vector space model

Probabilistic information retrieval

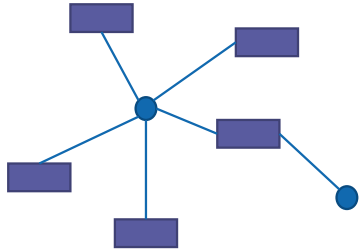
Language models

Alternate methodologies

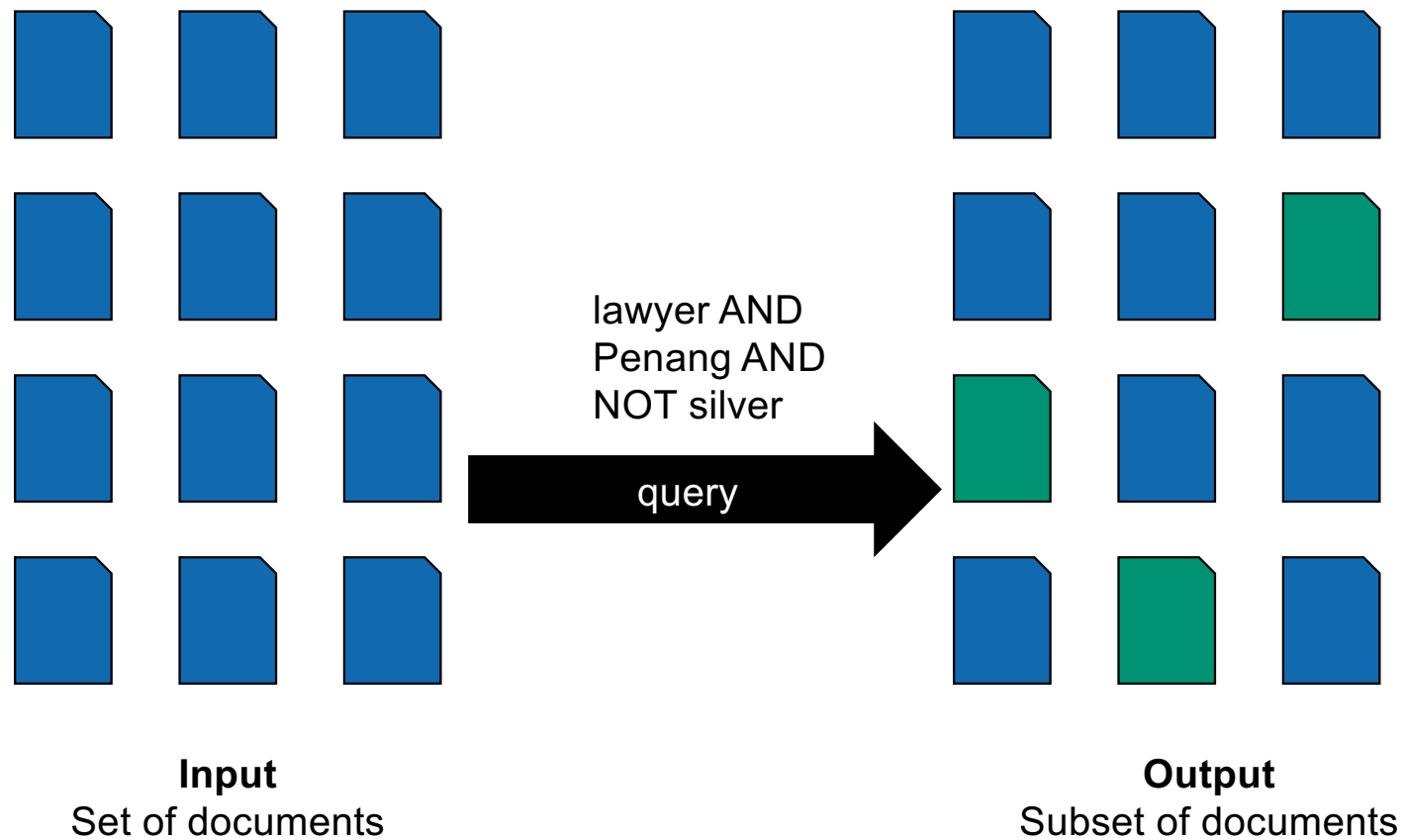
Data Shapes: Text

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam vel erat nec dui aliquet vulputate sed quis nulla. Donec eget ultricies magna, eu dignissim elit. Nullam sed urna nec nisl rhoncus ullamcorper placerat et enim. Integer varius ornare libero quis consequat. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Aenean eu efficitur orci. Aenean ac posuere tellus. Ut id commodo turpis.

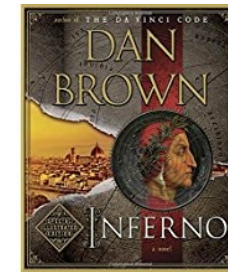
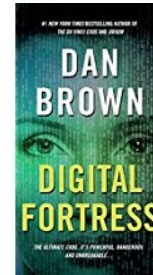
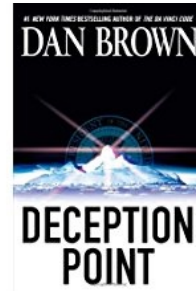
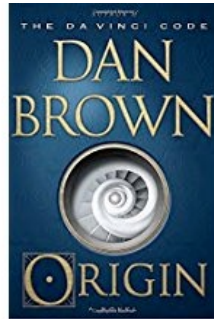
Praesent nec libero metus. Praesent at turpis placerat, congue ipsum eget, scelerisque justo. Ut volutpat, massa ac lacinia cursus, nisl dui volutpat arcu, quis interdum sapien turpis in tellus. Suspendisse potenti. Vestibulum pharetra justo massa, ac venenatis mi condimentum nec. Proin viverra tortor non orci suscipit rutrum. Phasellus sit amet euismod diam. Nullam convallis nunc sit amet diam suscipit dapibus. Integer porta hendrerit nunc. Quisque pharetra congue porta. Suspendisse vestibulum sed mi in euismod. Etiam a purus suscipit, accumsan nibh vel, posuere ipsum. Nulla nec tempor nibh, id venenatis lectus. Duis lobortis id urna eget tincidunt.



Boolean retrieval



Document

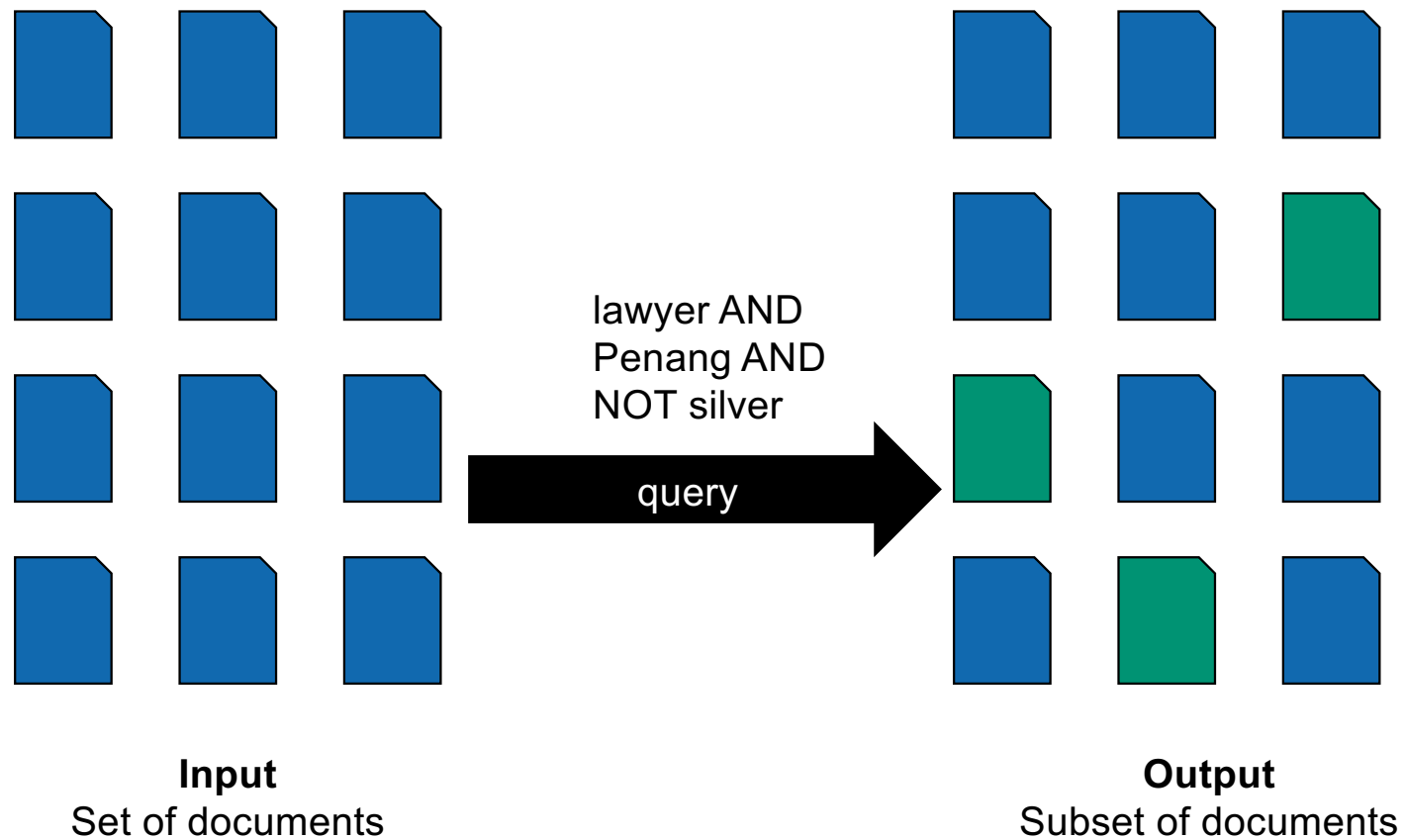


Documents

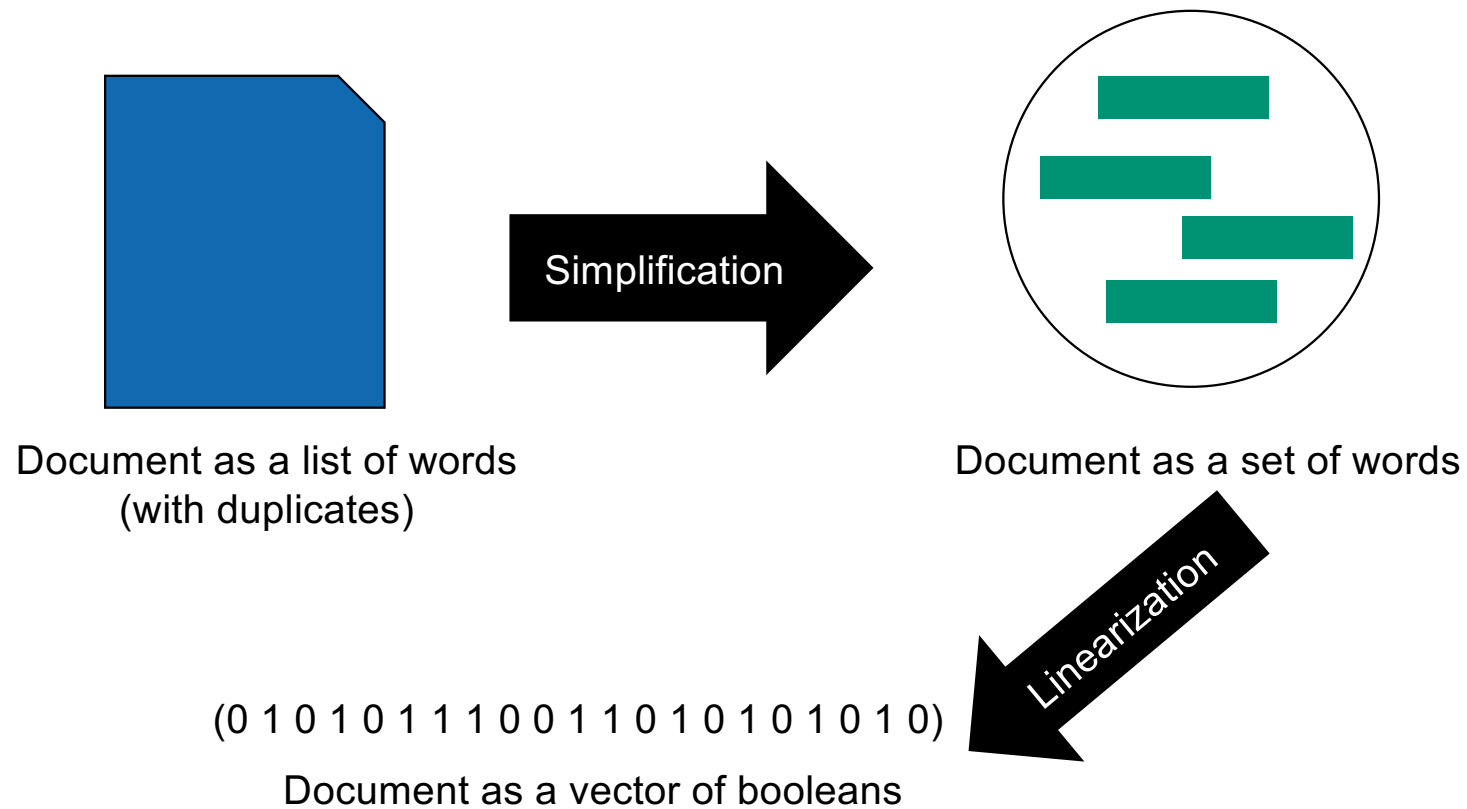
Term

Sherlock
lawyer
Switzerland
Unterwalden nid dem Wald
ETH Zürich
person
watch
run
paper
book
...

Boolean retrieval



Model and abstraction



Incidence Matrix

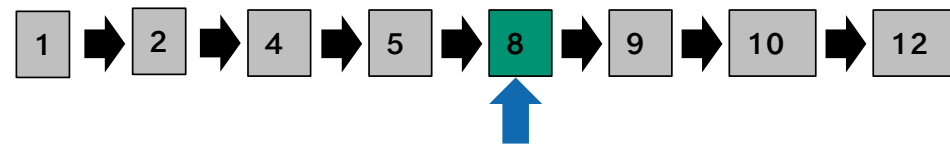
		Documents									
		1	2	3	4	5	6	7	8	9	10
Terms	t	1	0	1	1	1	0	0	0	1	1
	u	0	0	1	0	1	1	1	1	0	0
	v	0	1	1	1	0	1	0	1	0	1
	w	0	0	0	1	1	0	0	0	0	0
	x	1	0	1	1	1	0	1	0	0	1
	y	0	0	0	0	1	0	0	1	0	1

Warm up

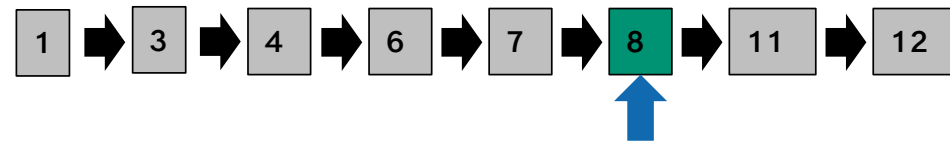
a	→	6	1	2	3	5	6	8
b	→	5	3	4	7	8	9	
c	→	5	1	2	4	5	7	
d	→	5	1	3	5	8	9	
e	→	4	2	3	4	7		
f	→	6	1	2	4	5	8	9
g	→	4	3	5	7	8		

Intersection algorithm

List A



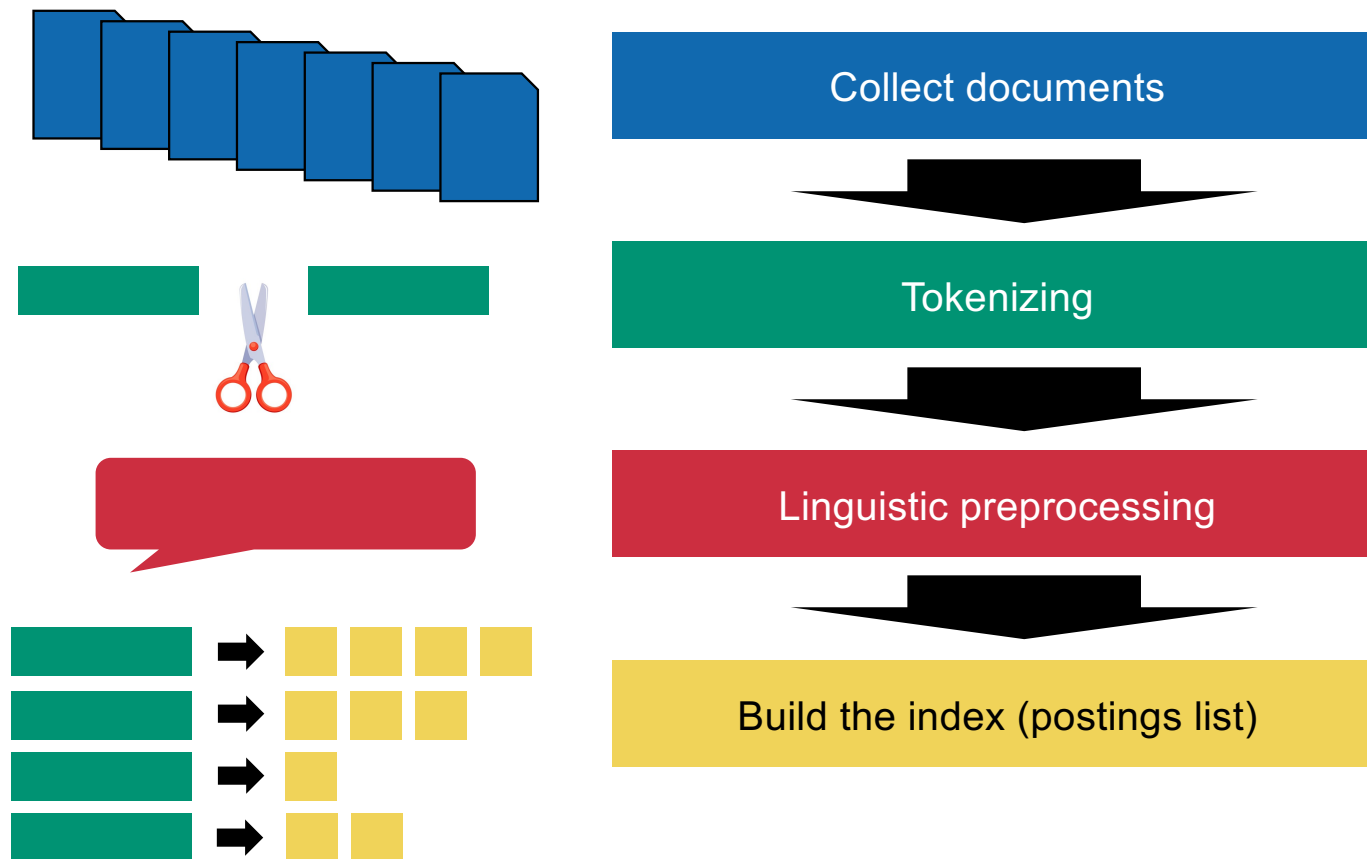
List B



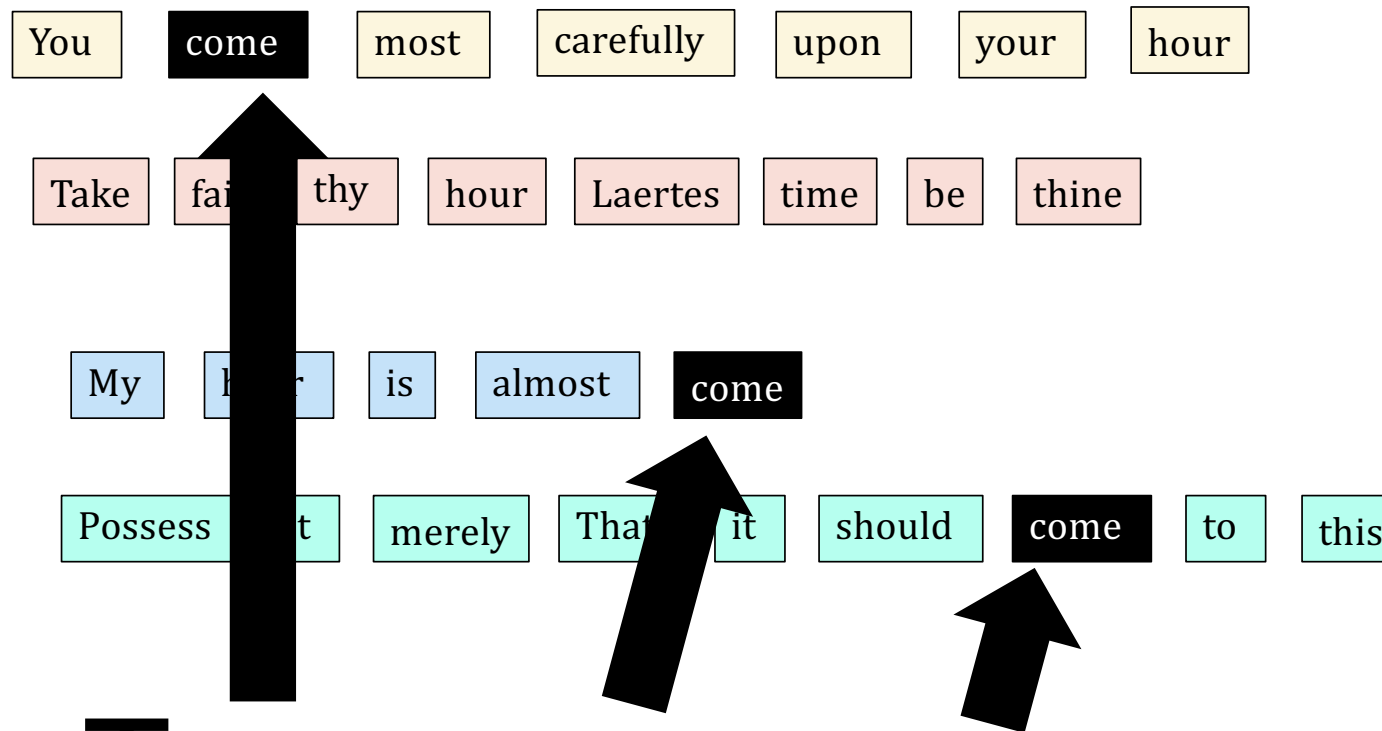
Intersection of A and B



Index construction



Type



Type = equivalence class (same sequences)

Stop words

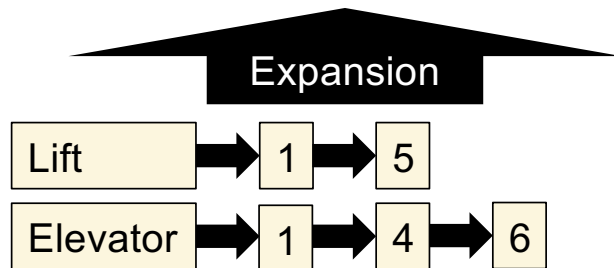
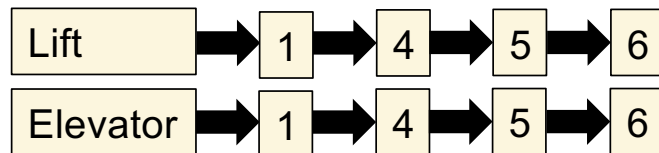
a
an
and
are
as
at
be
by
for
from
has
he
in

is
it
its
of
on
that
the
to
was
were
will
with

Query expansion

Upon indexing

Lift |

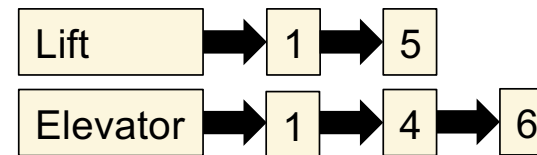


Upon querying

Lift |



Lift OR Elevator

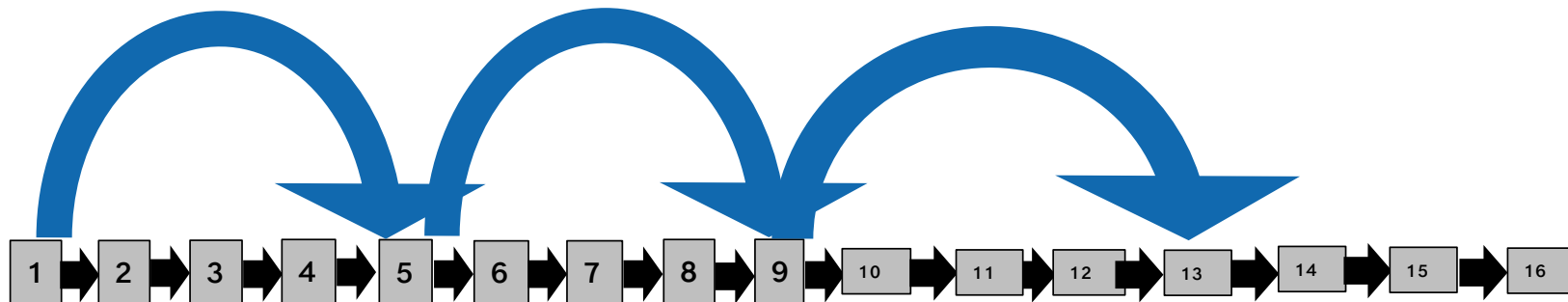


Porter Stemmer

(m>0) ENCI -> ENCE	valenci -> valence
(m>0) ANCI -> ANCE	hesitanci -> hesitance
(m>0) IZER -> IZE	digitizer -> digitize
(m>0) ABLI -> ABLE	conformabli -> conformable
(m>0) ALLI -> AL	radicalli -> radical
(m>0) ENTLI -> ENT	differentli -> different
(m>0) ELI -> E	vileli -> vile
(m>0) OUSLI -> OUS	analogousli -> analogous
(m>0) IZATION -> IZE	vietnamization -> vietnamize
(m>0) ATION -> ATE	predication -> predicate
(m>0) ATOR -> ATE	operator -> operate
(m>0) ALISM -> AL	feudalism -> feudal
(m>0) IVENESS -> IVE	decisiveness -> decisive
(m>0) FULNESS -> FUL	hopefulness -> hopeful
(m>0) OUSNESS -> OUS	callousness -> callous

<https://tartarus.org/martin/PorterStemmer/>

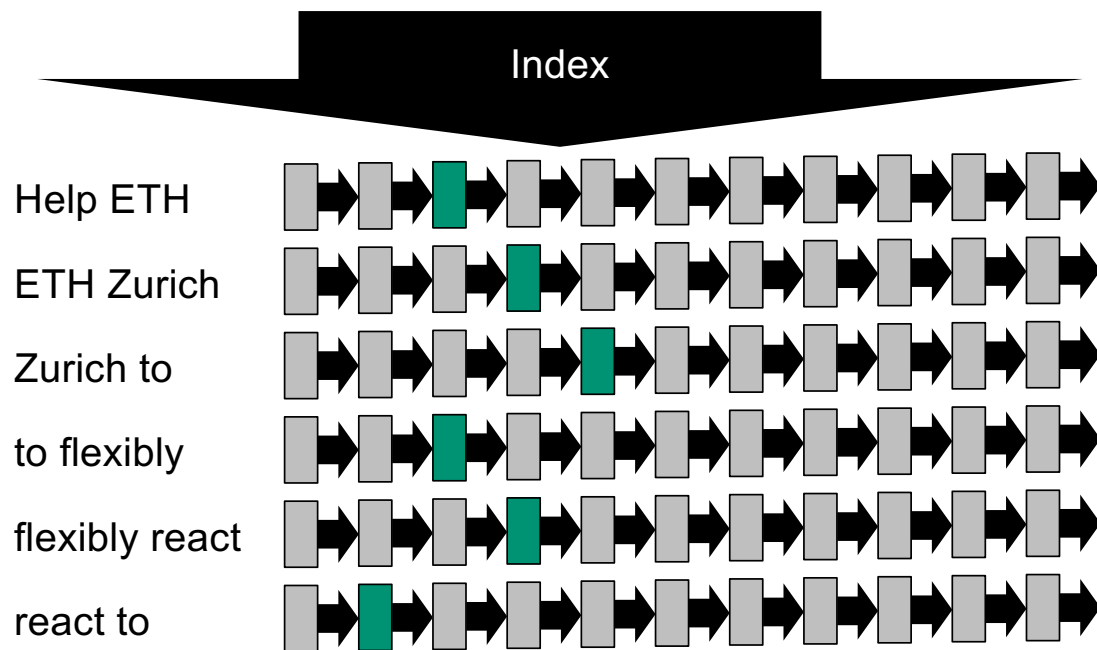
Skip lists



In practice $\sqrt{\text{Number of postings}}$

Bi-word indices (Phrase search feature)

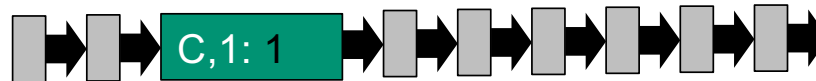
Help ETH Zurich to flexibly react to new challenges and to set new accents in the future.



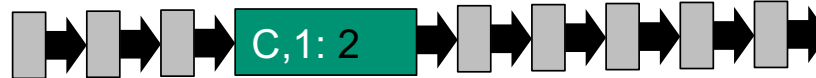
Positional index (phrase search feature)

"ETH Zurich"

Help



ETH



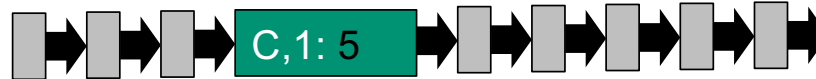
Zurich



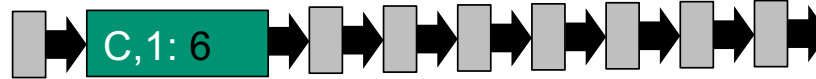
to



flexibly

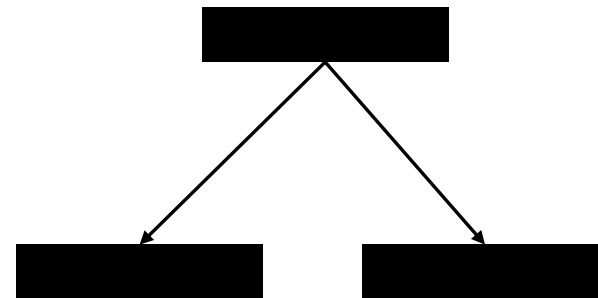


react



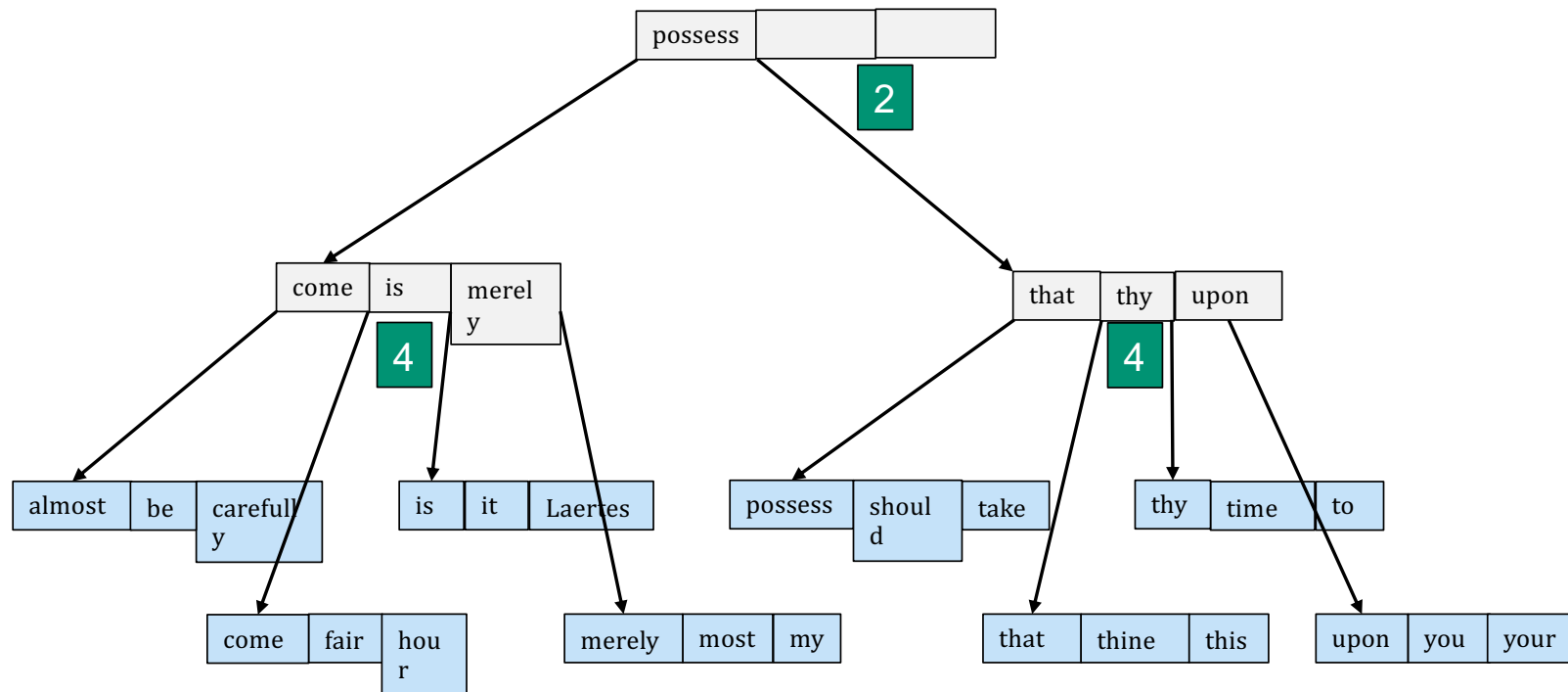
Search structures

Hash tables



Trees (B, B+)

B+-tree



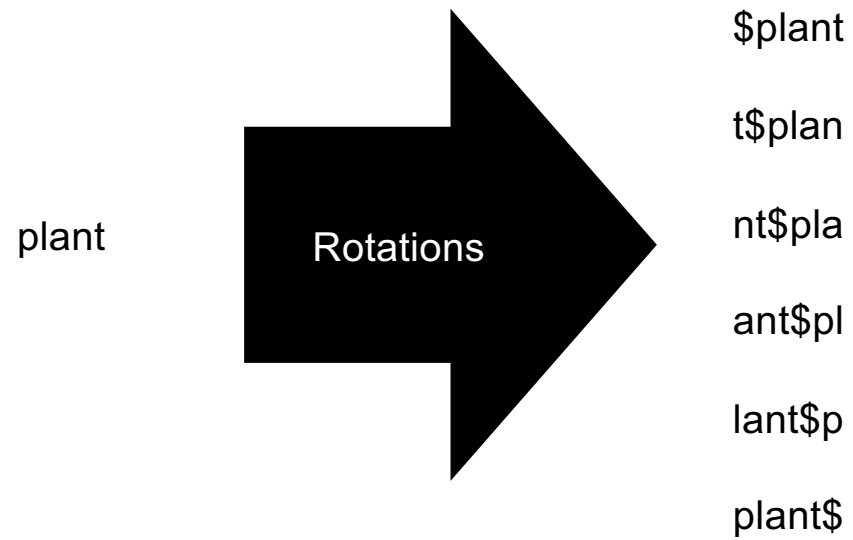
✓ But it's fine if the root has less.

Wildcard queries

foo*eth*bar

multiple wildcards

Permuterm index



k-grams

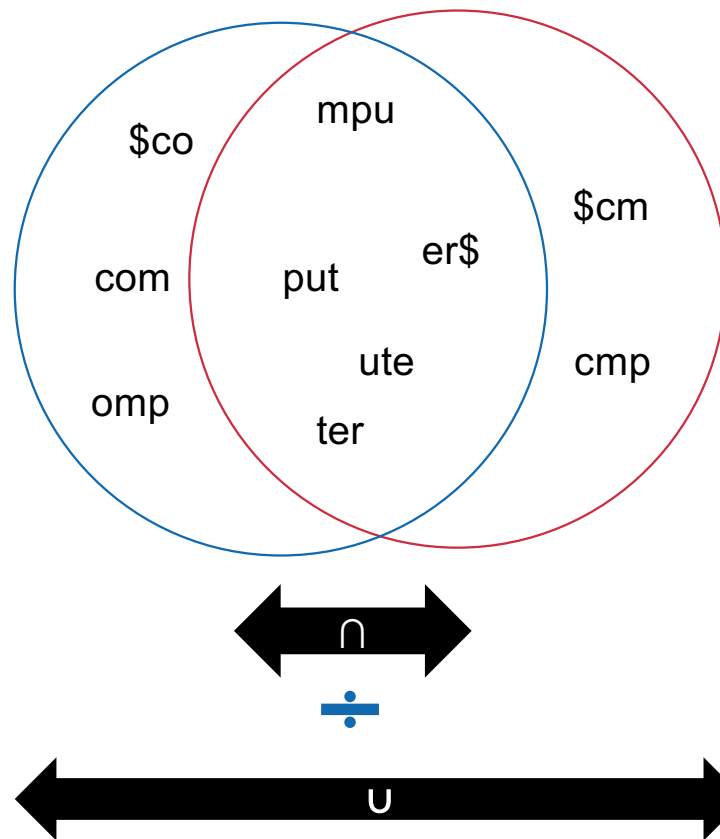
computer	1-grams	\$, c, o, m, p, u, t, e, r, \$	Not very useful
	2-grams	\$c, co, om, mp, pu, ut, te, er, r\$	Usable zone
	3-grams	\$co, com, omp, mpu, put, ute, ter, er\$	
	4-grams	\$com, comp, ompu, mput, pute, uter, ter\$	
	5-grams	\$comp, compu, omput, mpute, puter, uter\$	Not space efficient
	6-grams	\$compu, comput, ompute, mputer, puter\$	
	7-grams	\$comput, compute, omputer, mputer\$	
	...		

Edit distance

	#	a	t	e
#	0	1	2	3
c	1	1	2	3
a	2	1	2	3
t	3	2	1	2

The diagram illustrates the edit distance between the strings 'c' and 'ate'. The table shows the edit distance for each character in 'c' against each character in 'ate'. The sequence of edits is shown by arrows: a vertical arrow from 'c' to 'a', a diagonal arrow from 'a' to 't', a horizontal arrow from 't' to 'e', and a diagonal arrow from 'a' to 'e'. The final result 'ate' is highlighted in a yellow box.

Jaccard coefficient

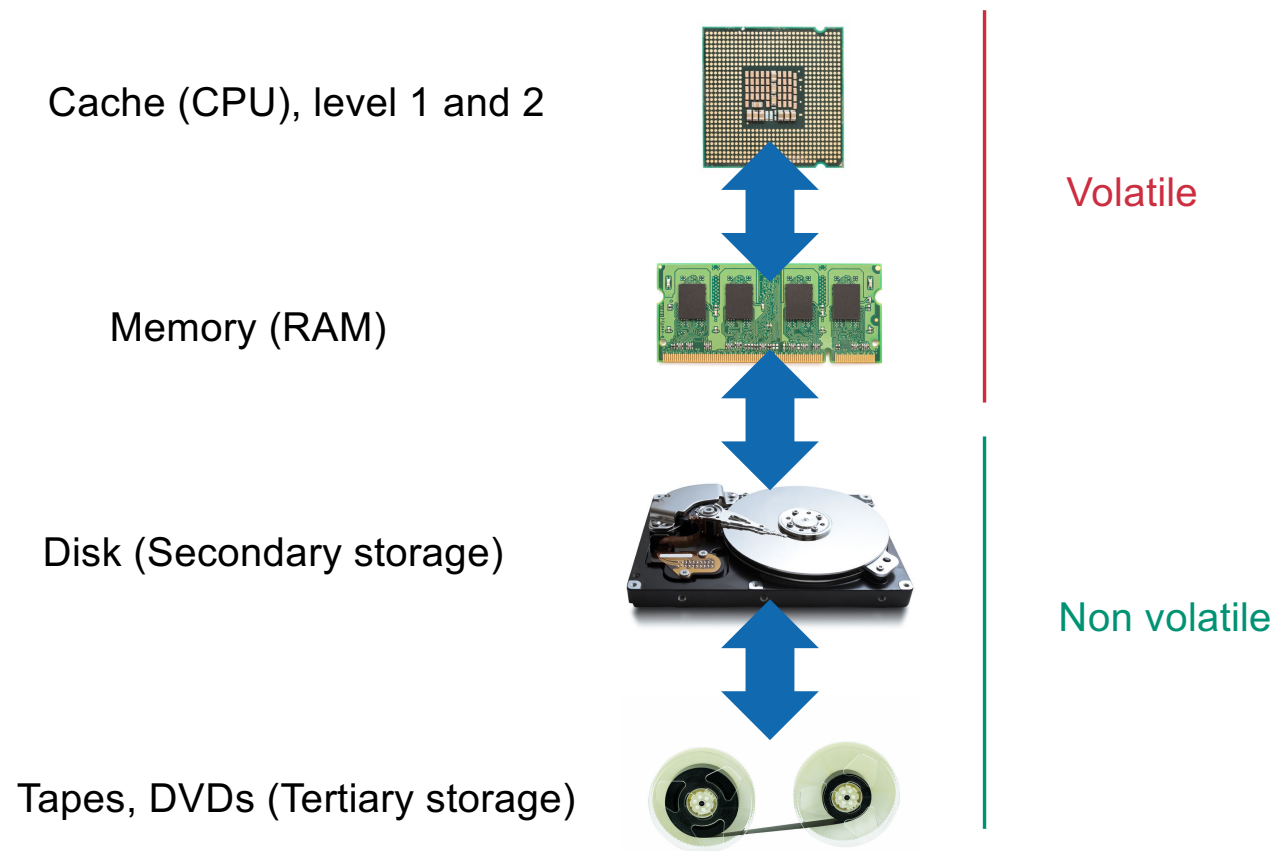


$$= 5 / 10 = 0.5$$

Soundex algorithm

Change...	To...
A E H I O U W Y	0
B F P V	1
C G J K Q S X Z	2
D T	3
L	4
M N	5
R	6

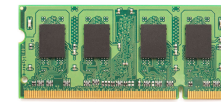
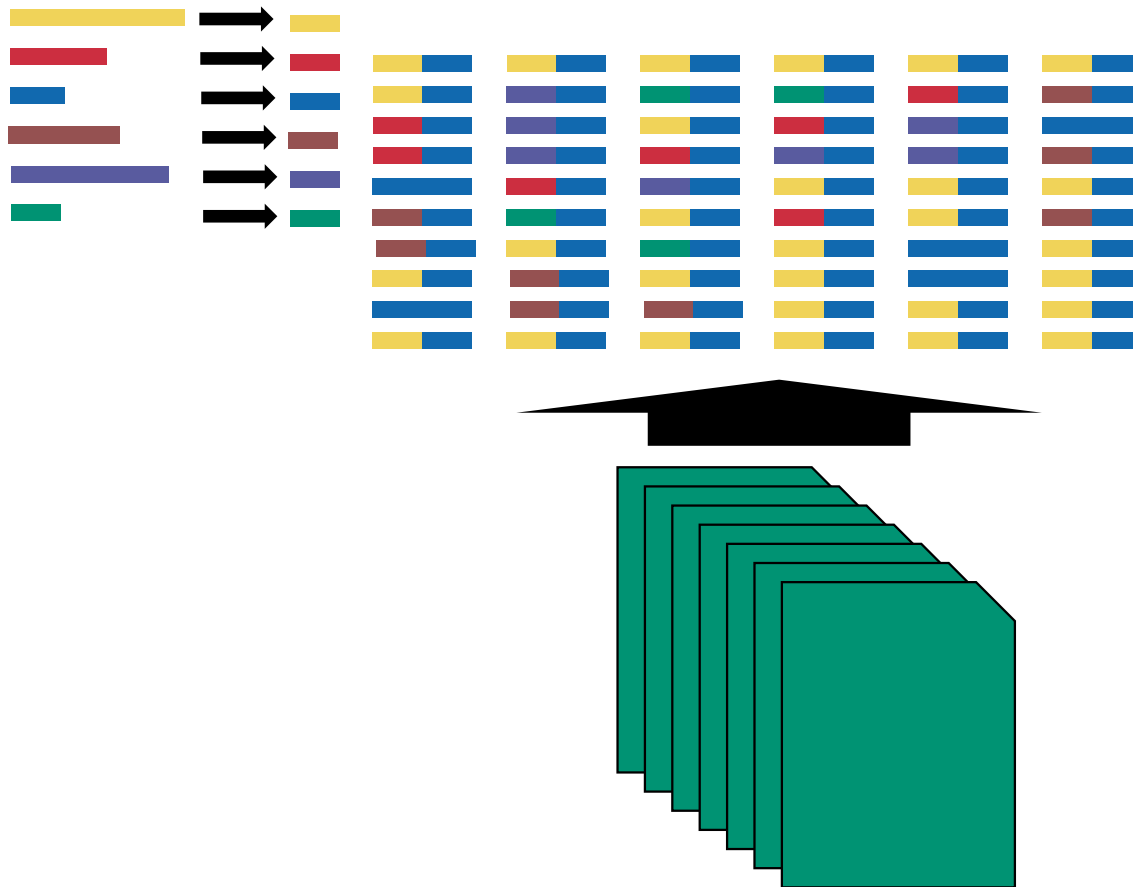
Memory hierarchy



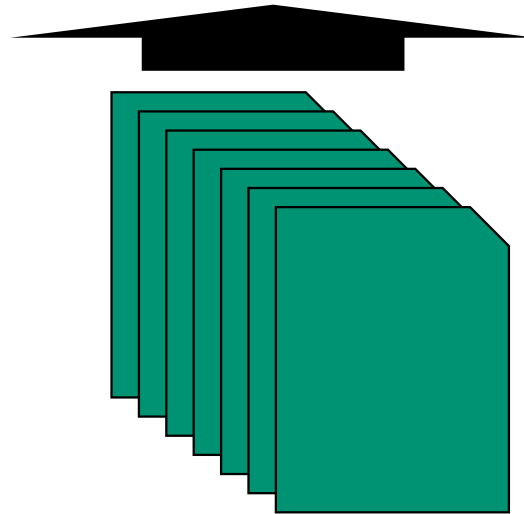
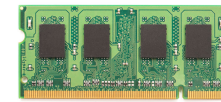
TermIDs

t1	1	t1	2	t1	3	...
t2	3	t2	4	t2	7	...
t3	1	t3	2	t3	4	...
t4	1	t4	3	t4	5	...
t5	2	t5	3	t5	4	...
t6	1	t6	2	t6	4	...
t7	3	t7	5	t7	7	...

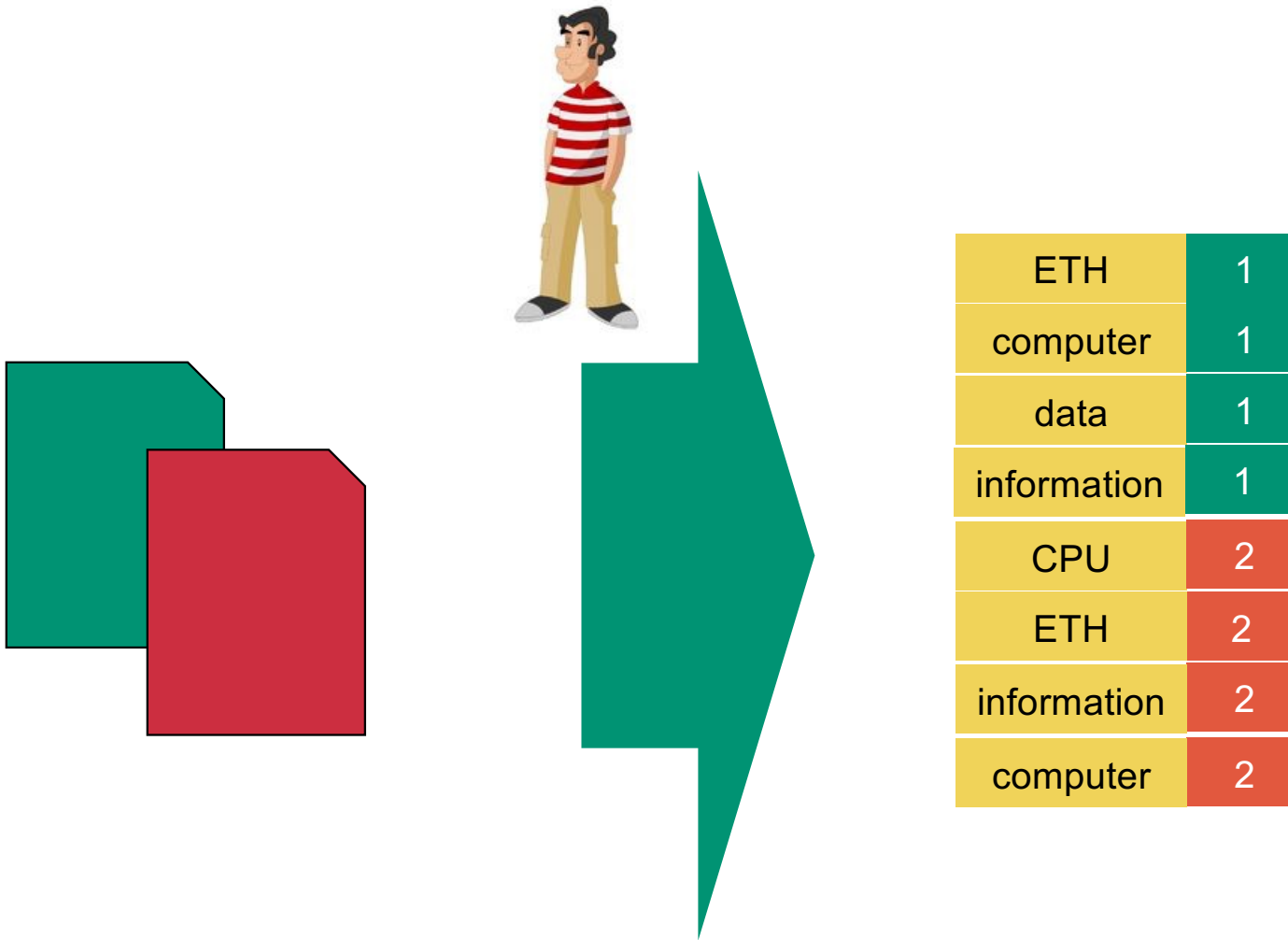
Blocked Sort-Based Indexing



Single-Pass In-Memory Indexing

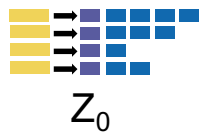


MapReduce

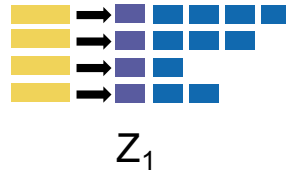


Logarithmic Merging

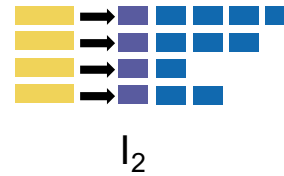
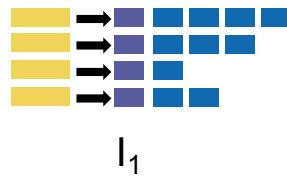
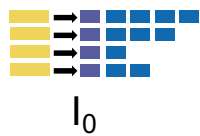
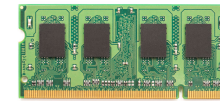
n postings



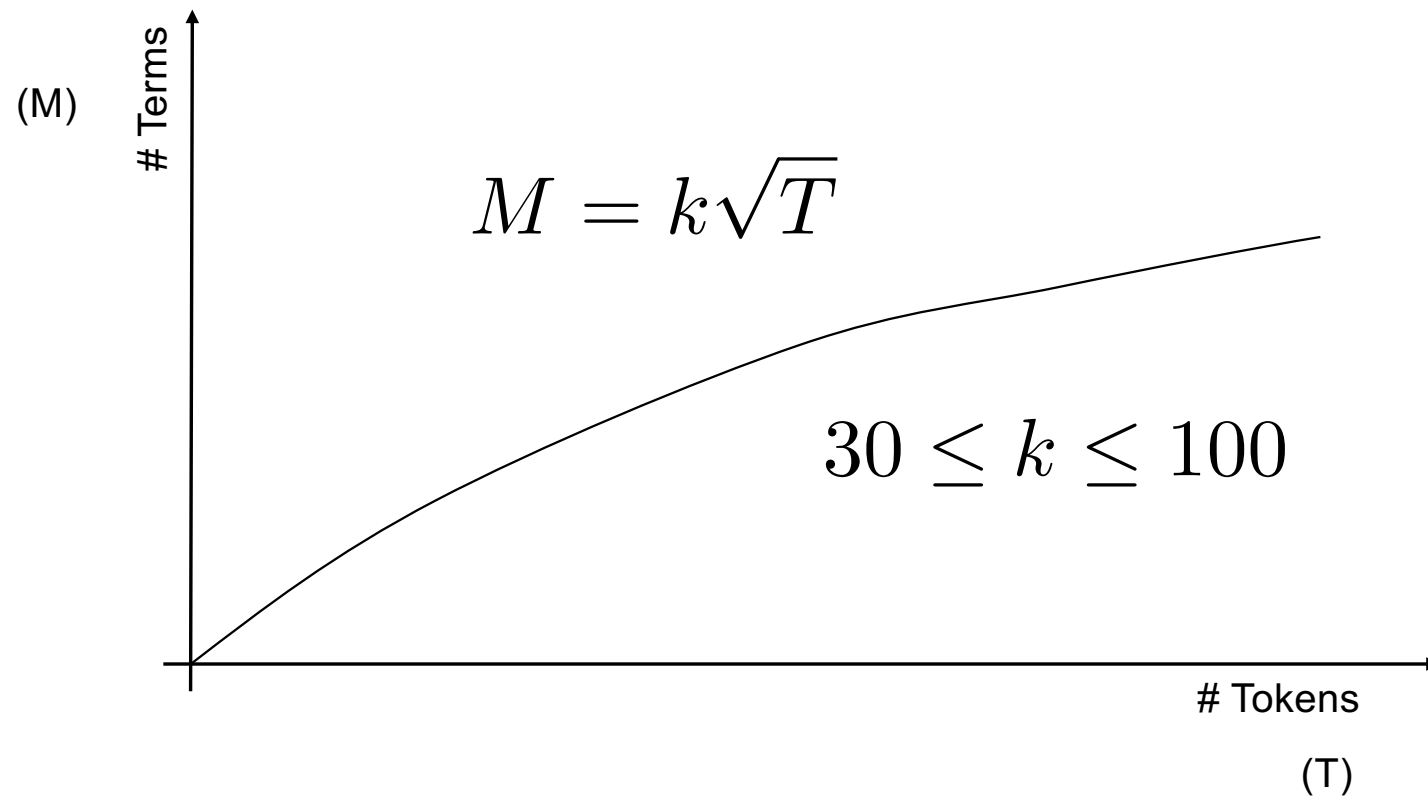
$2n$ postings



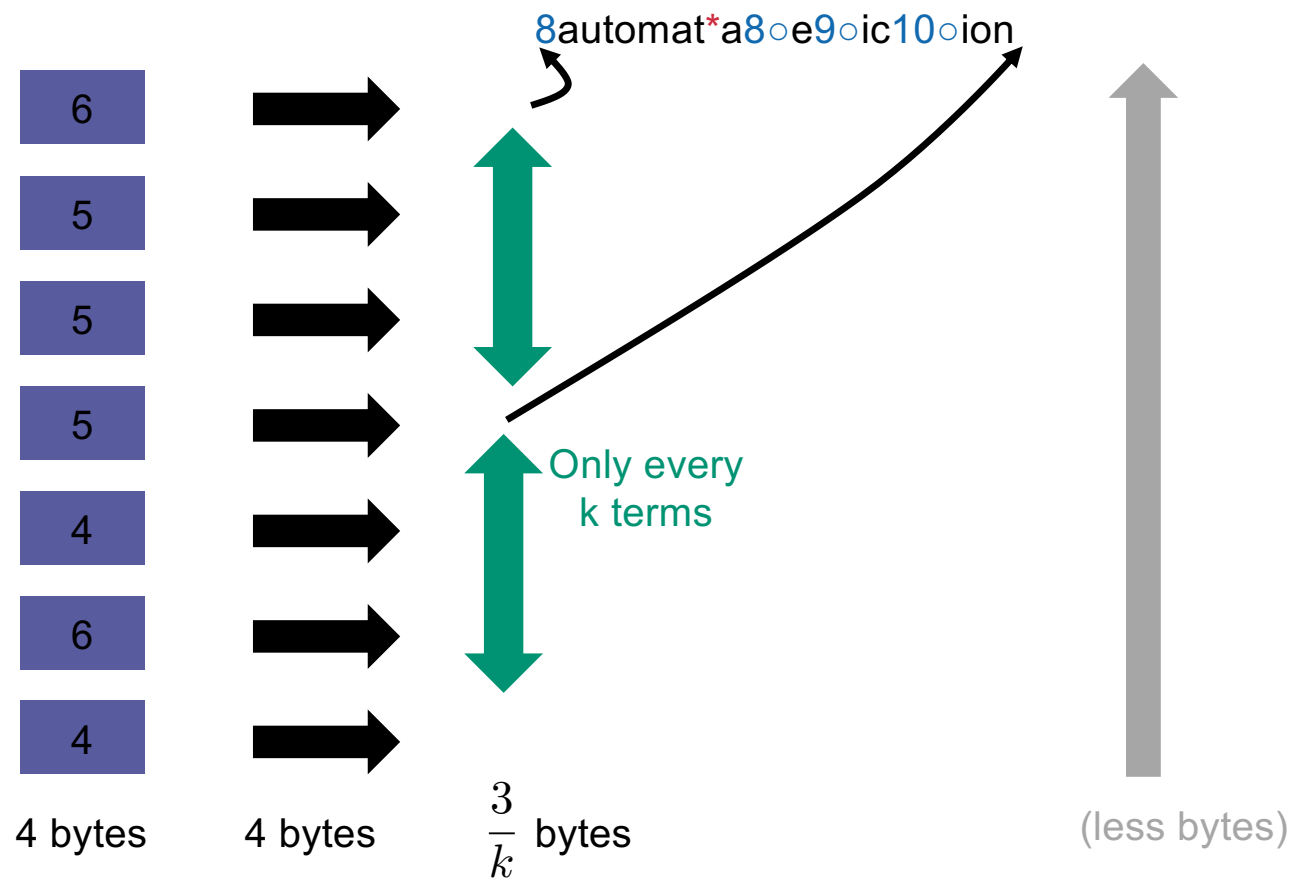
$4n$ postings



Heap's law



Compression: Front coding

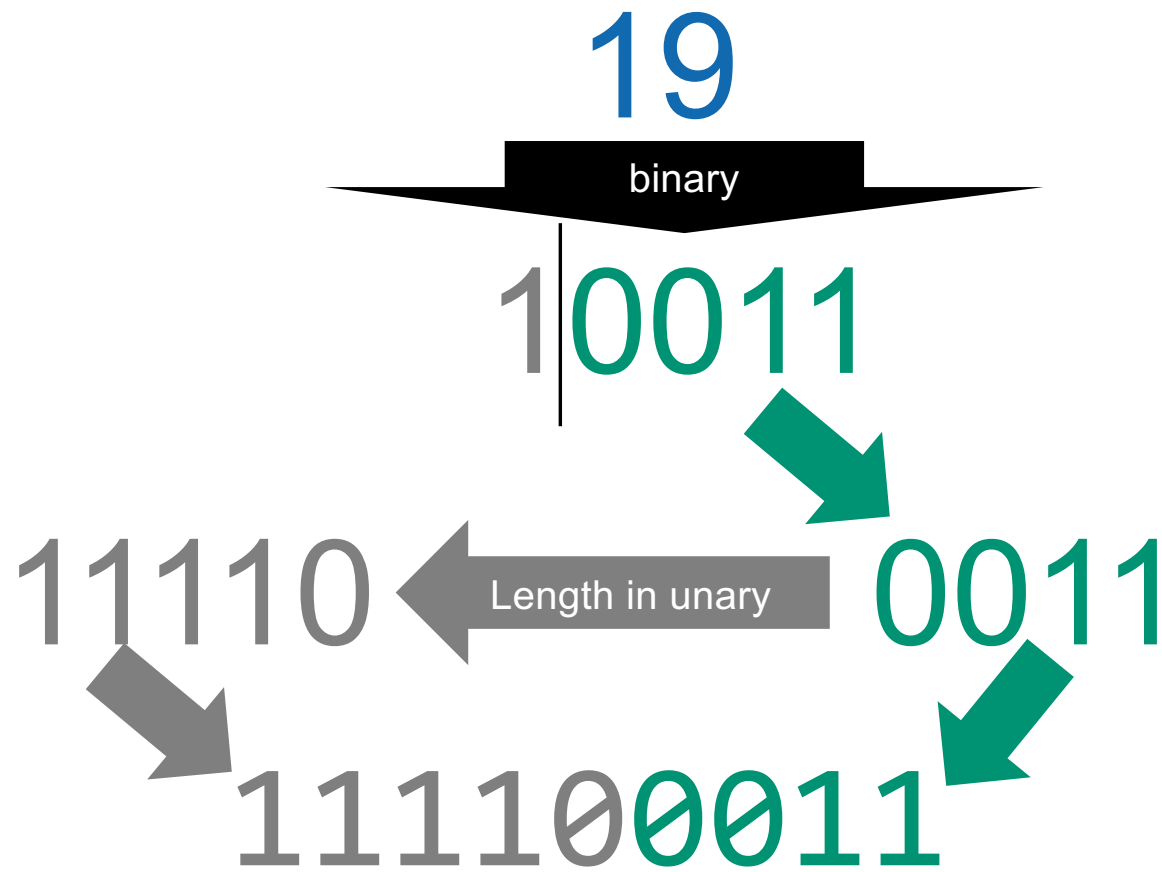


Variable byte encoding

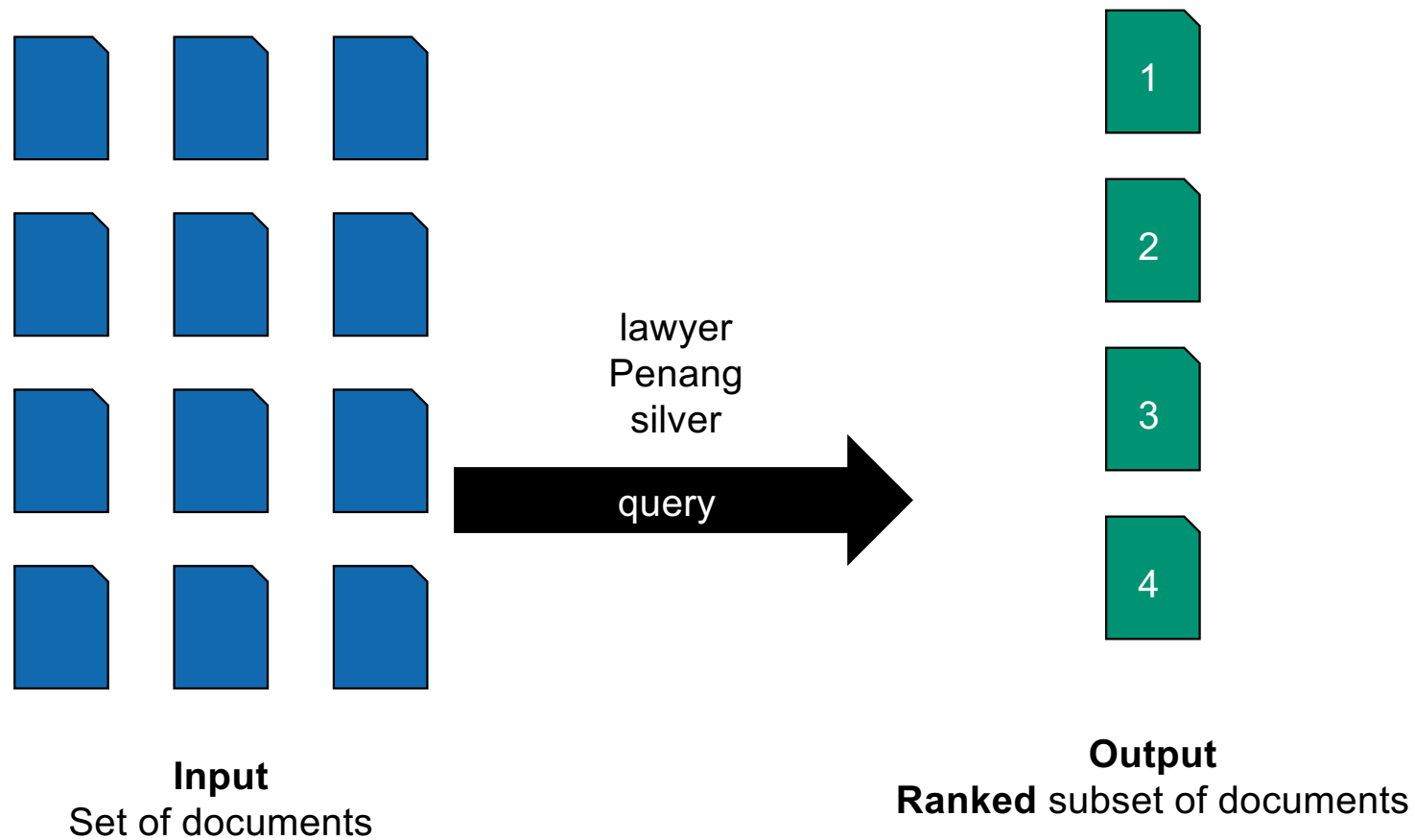
decimal	binary	variable byte encoding
0	0	0000
1	1	0001
2	10	0010
3	11	0011
4	100	0100
5	101	0101
6	110	0110
7	111	0111
8	1000	1001 0000
9	1001	1001 0001
10	1010	1001 0010
11	1011	1001 0011
12	1100	1001 0100
13	1101	1001 0101
14	1110	1001 0110
15	1111	1001 0111
16	10000	1010 0000
17	10001	1010 0001
18	10010	1010 0010
19	10011	1010 0011
20	10100	1010 0100
21	10101	1010 0101
22	10110	1010 0110
23	10111	1010 0111
...	...	...
64	1000000	1001 1000 0000

50%
less space

Gamma encoding



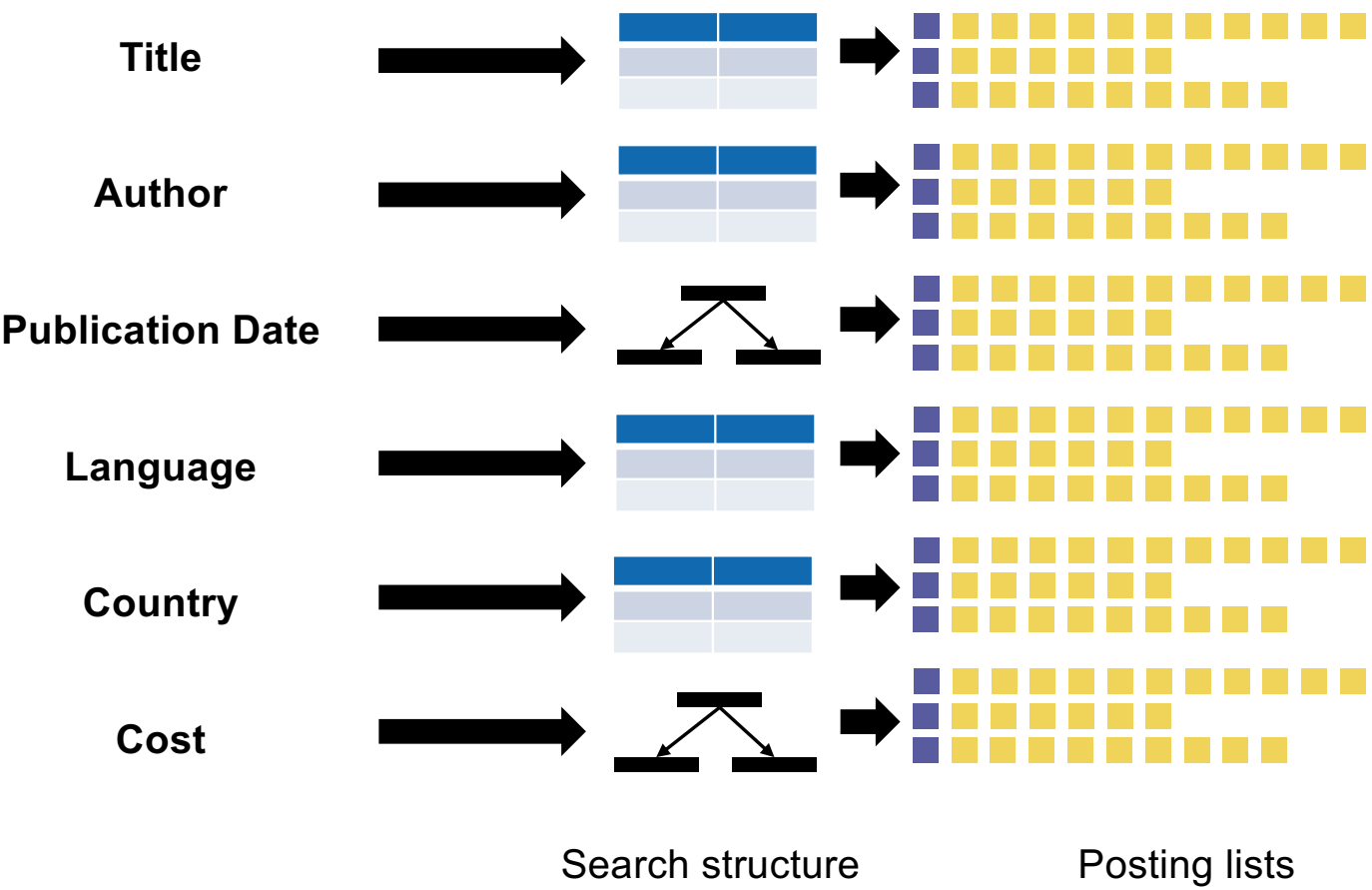
Ranked retrieval



Parametric search

Title	Algorithms		
Author			
Publication Date			
Language			
Country			
Cost		to	
	\$		\$
	Search		

Parametric indices



Term frequency, (Inverted) Document frequency,

tf	A	B
foo	5	1
bar	0	4
foobar	2	1

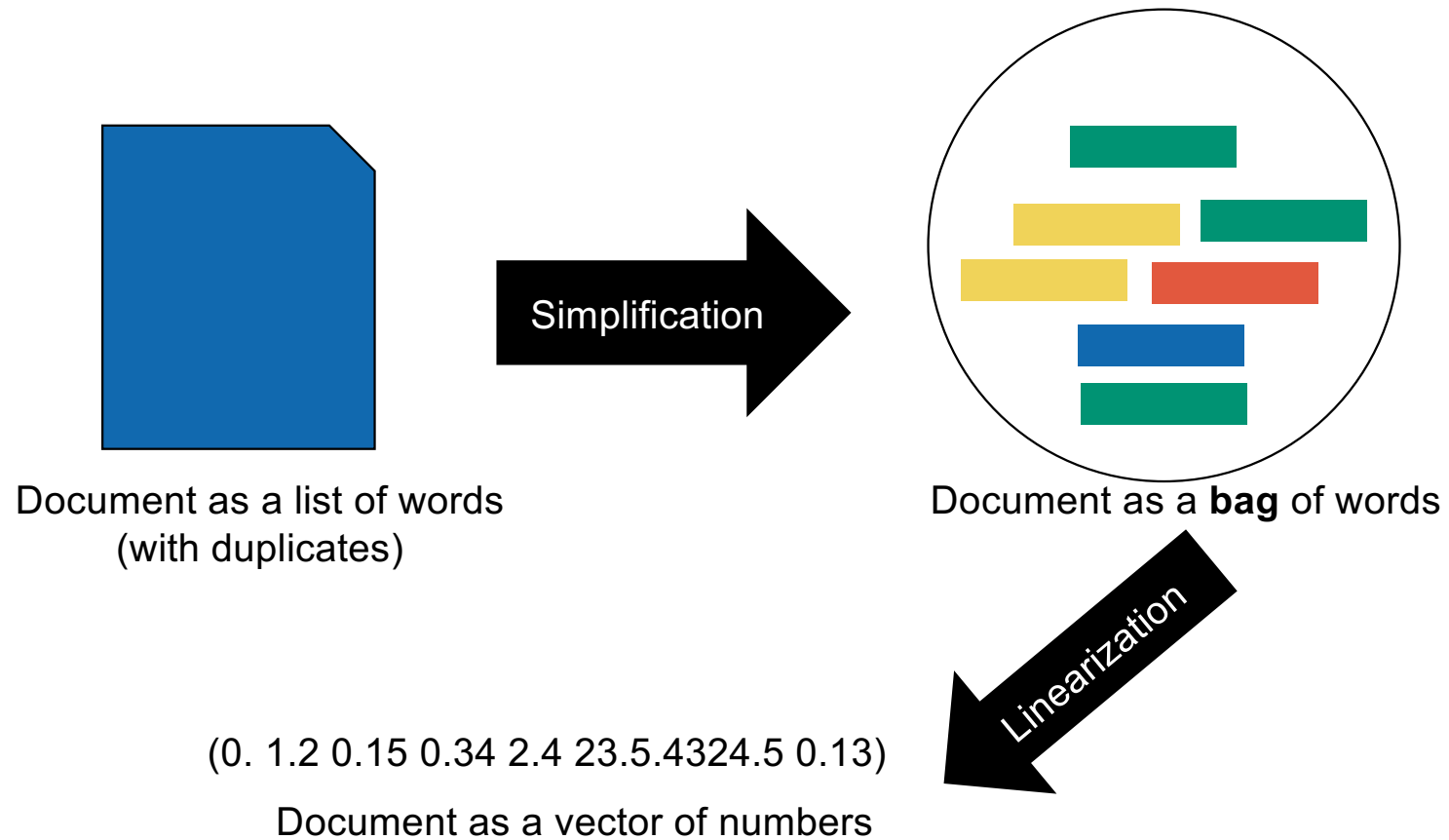


tf-idf	A	B
foo	25	5
bar	0	40
foobar	6	3



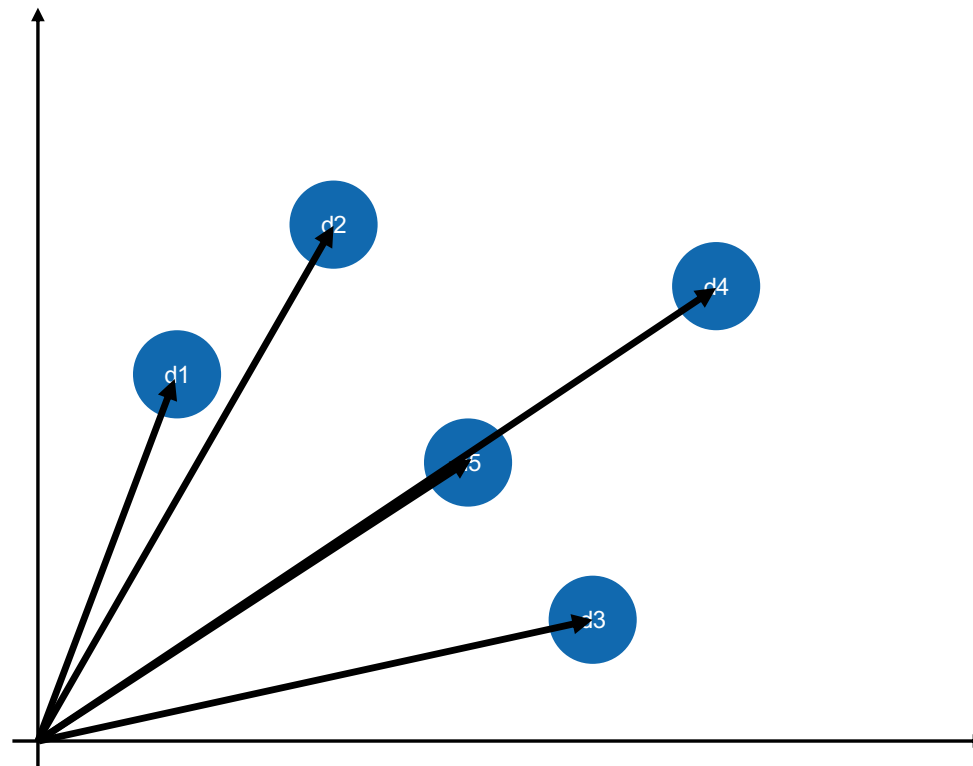
idf	
foo	5
bar	10
foobar	3

Model and abstraction



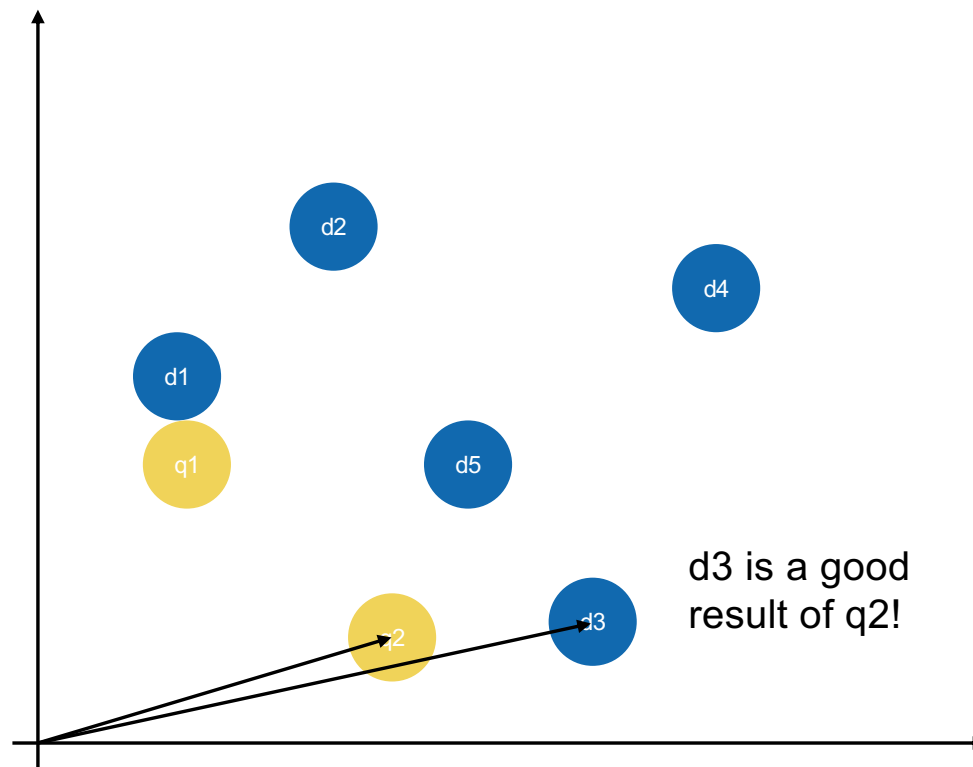
Vector-Space Model

Documents
= vectors in the
first quadrant
of
 $\mathbb{R}^M$

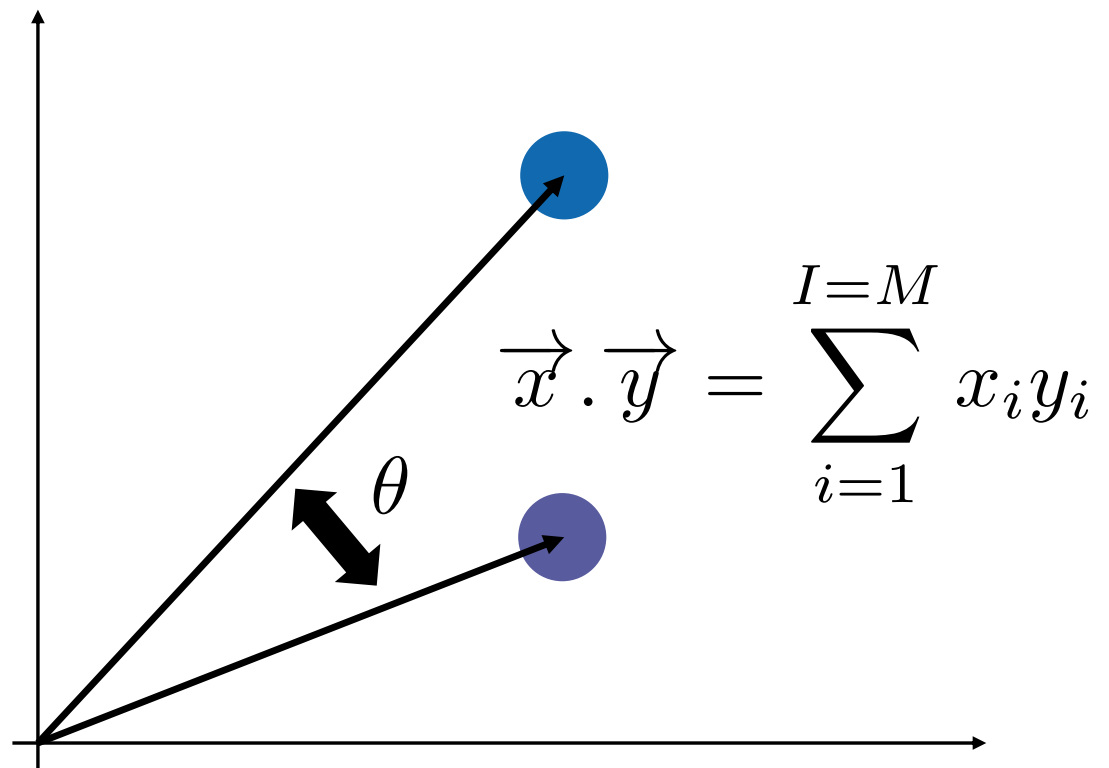


Queries as vectors

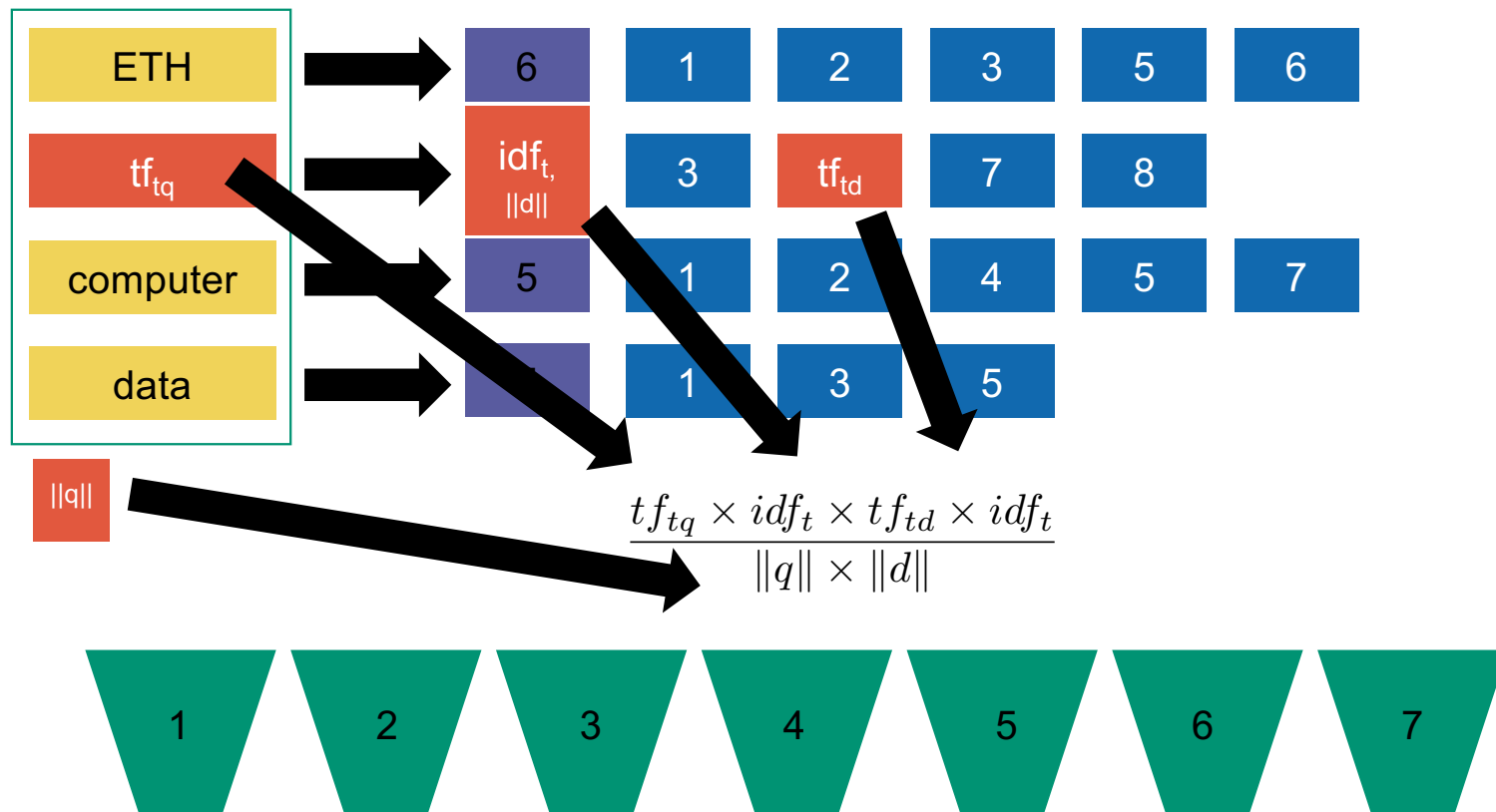
Queries
= points in the
first quadrant
of
 $\mathbb{R}^M$



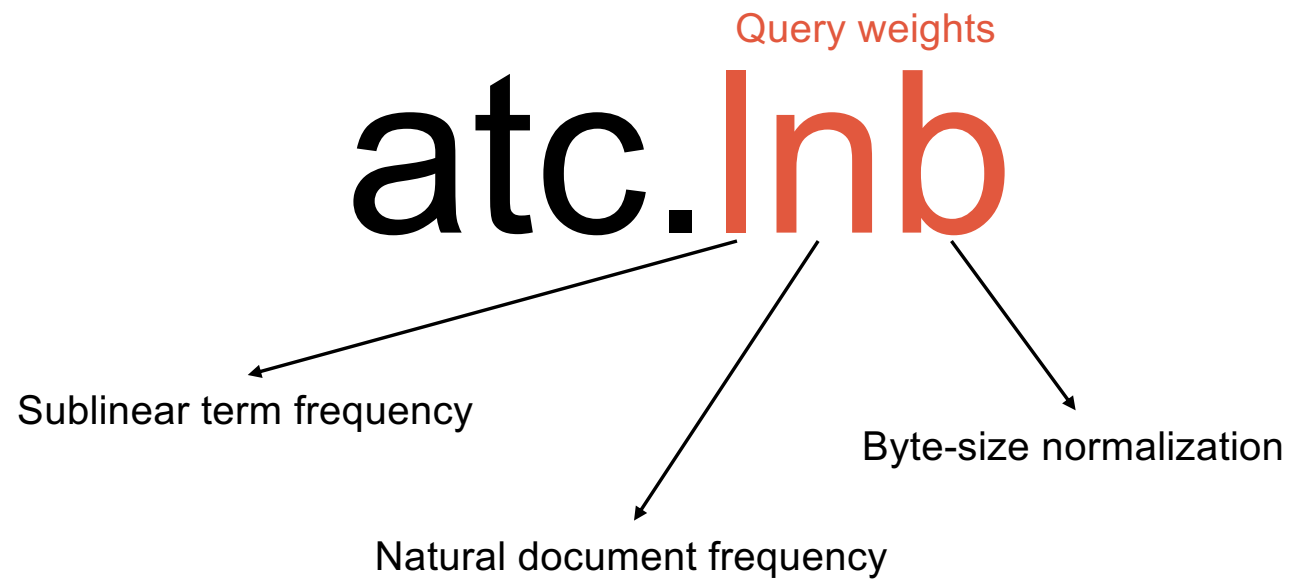
Inner product as score



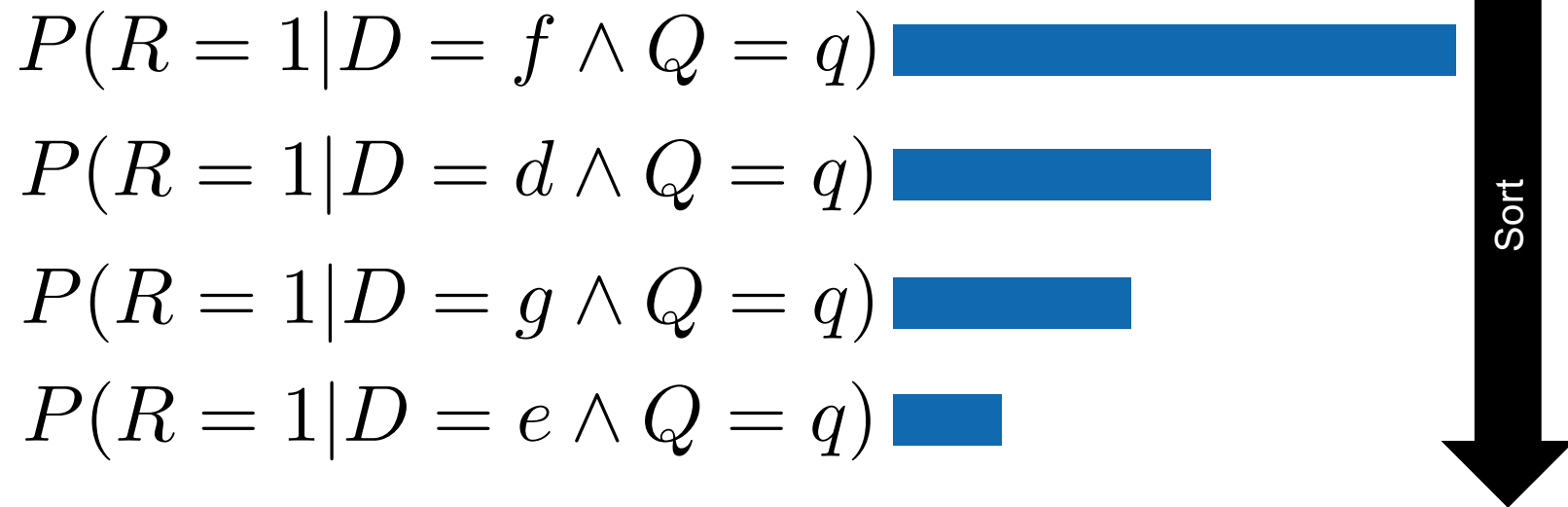
Evidence accumulation



SMART notation



Probabilistic Information Retrieval



... falling back to Ranked Retrieval and evidence accumulation!

$$RSV_d = \sum_{k|d_k=1 \wedge q_k=1} \log \frac{N}{df_t}$$

This justifies idf weighting in the Vector-Space Model!

Language models and document generation

$$\begin{aligned} P(D = (d_1, d_2, d_3, d_4)) = & \\ & (1 - p_{stop})P(D_1 = d_1) \\ & (1 - p_{stop})P(D_2 = d_2|D_1 = d_1) \\ & (1 - p_{stop})P(D_3 = d_3|D_1 = d_1 \wedge D_2 = d_2) \\ & (1 - p_{stop})P(D_4 = d_4|D_2 = d_2 \wedge D_3 = d_3)p_{stop} \end{aligned}$$

Language models

Enters a query q



Thought experiment: imagine that:

- we picked a random document and built its model
- we used this model to generate a new document
- that document turns out to be q

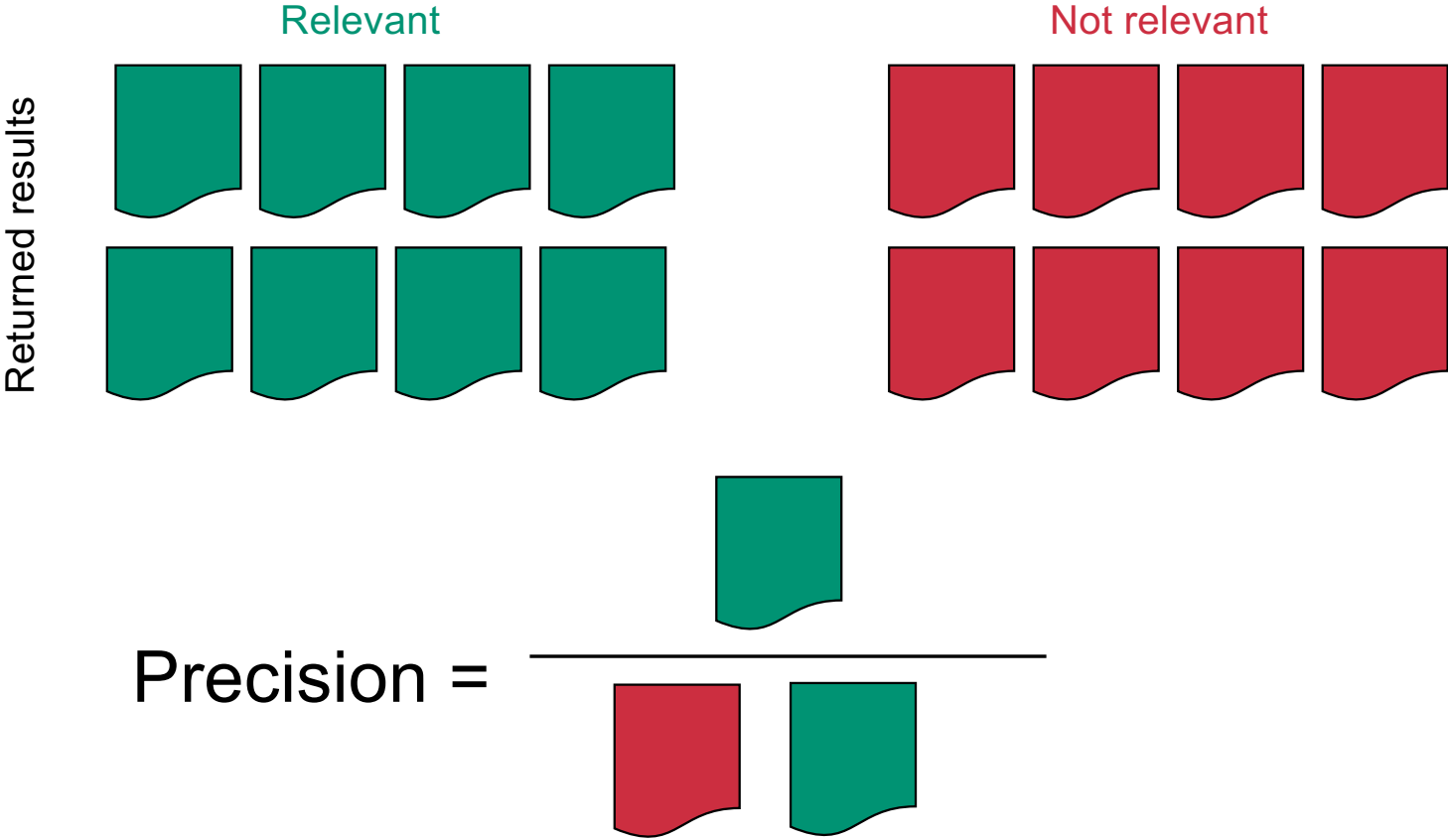
What document is the **most likely** to have been picked and to have generated q ?

Unordered unigram models and multinomials

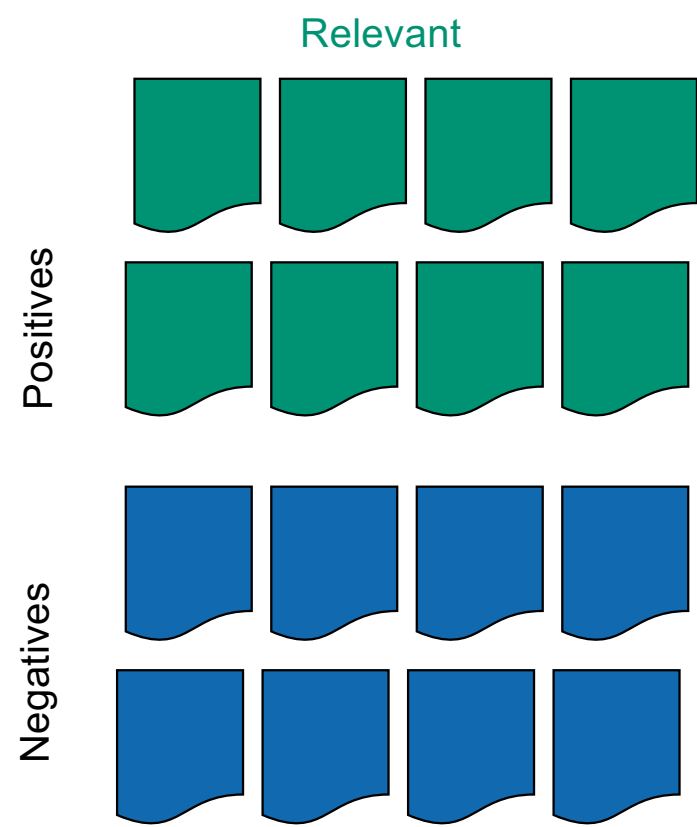
$$P(Q = q | D = d) = \frac{L_q!}{\prod_t t f_{t,q}!} \prod_t \left(\frac{t f_{t,d}}{L_d} \right)^{t f_{t,q}} (1 - p_{stop})^{L_q} p_{stop}$$

This justifies tf weighting in the Vector-Space Model!

Results

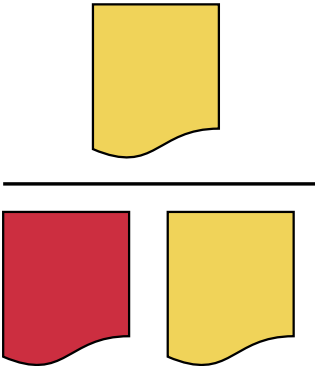


Results

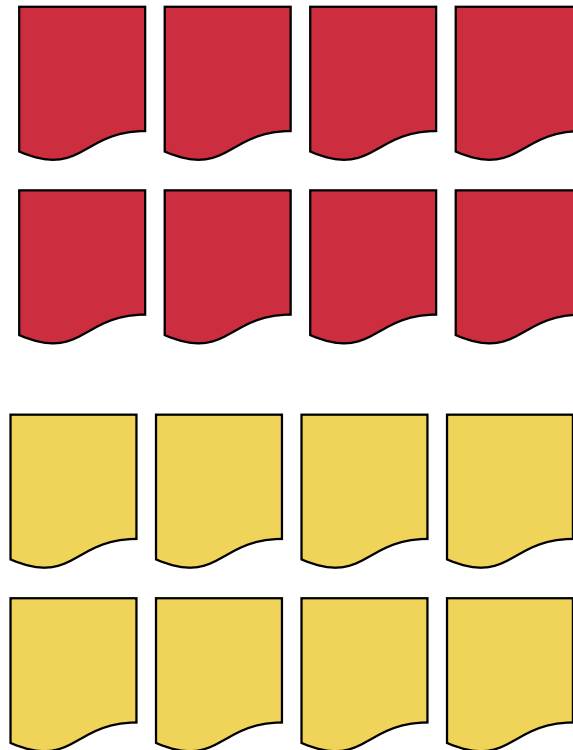


Recall = $\frac{\text{Green rectangles above the line}}{\text{Green rectangles above the line} + \text{Green rectangle below the line}}$

Specificity

$$\text{Specificity} = \frac{\text{True Negative}}{\text{True Negative} + \text{False Positive}}$$


Not relevant



F measure: harmonic mean

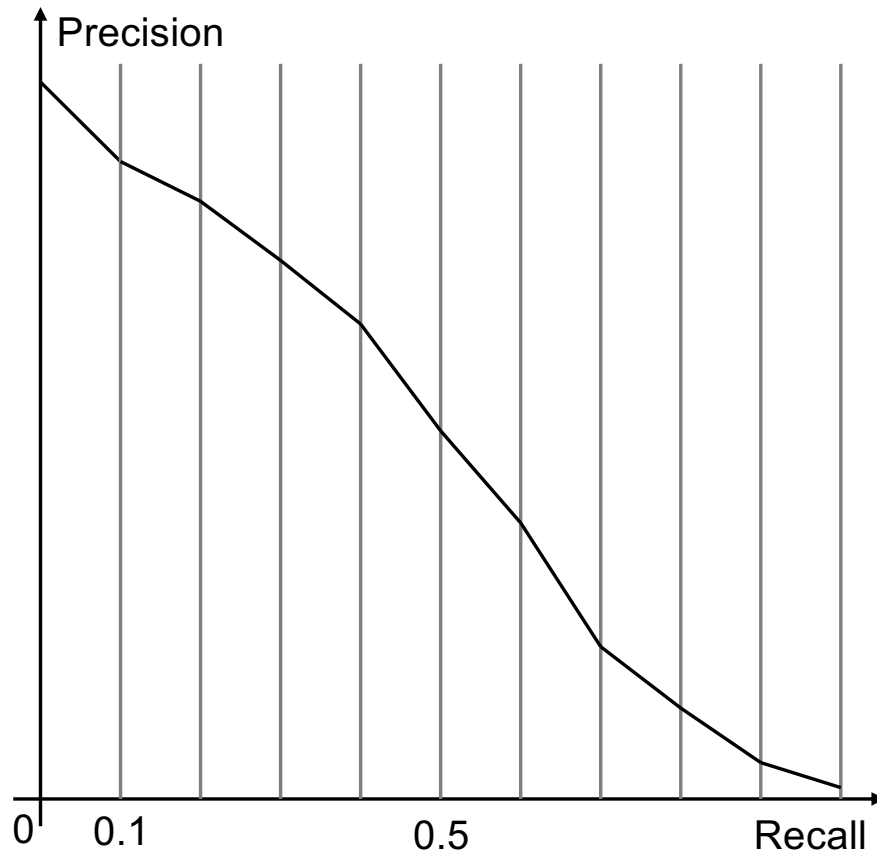
$$F_{\alpha} = \frac{1}{\frac{\alpha}{P} + \frac{1-\alpha}{R}}$$

$$\alpha = 0$$

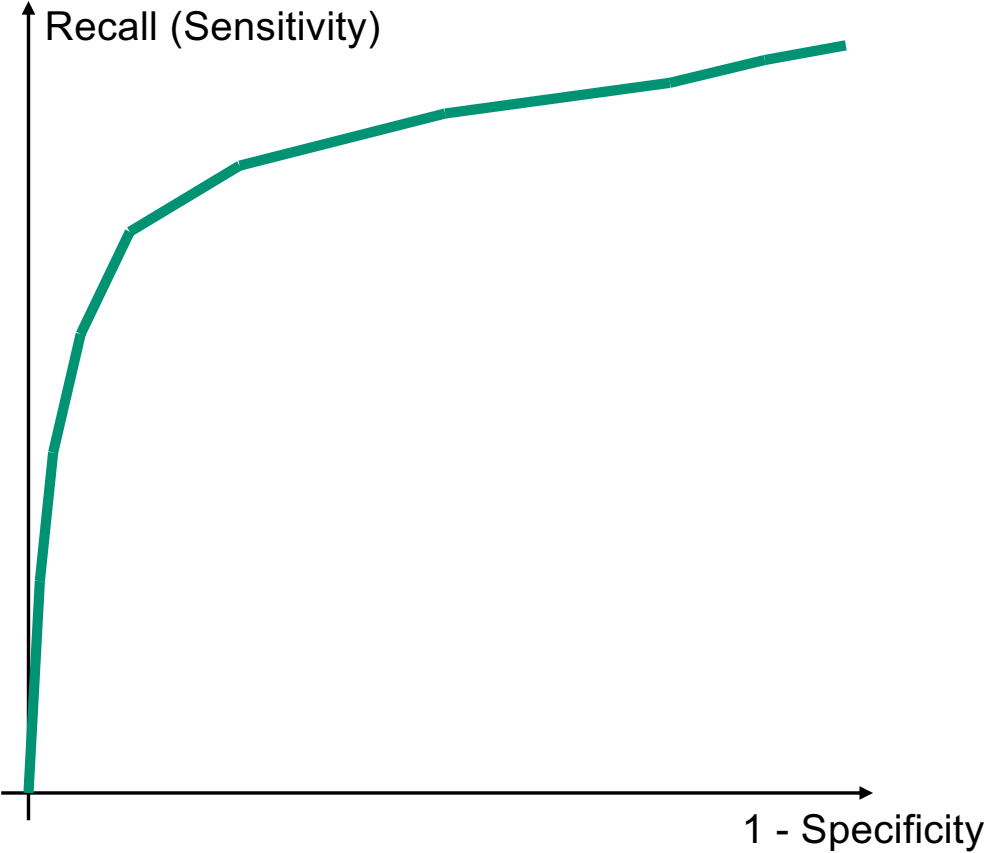
$$\alpha = 1$$



Precision-Recall curves



ROC Curves



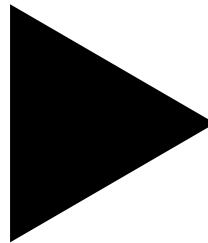


Exam

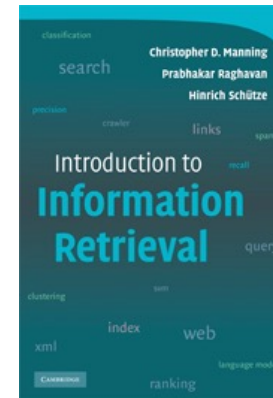
Material



Slides



Recordings



Textbook



Theoretical exercises



Practical exercises



Clicker questions

Let's start training on the exam!

